19 Management of Endodontic Emergencies

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Emergency Classifications

The proper diagnosis and effective management of acute dental pain are possibly the most rewarding and satisfying aspects of providing dental care. An endodontic emergency is defined as pain or swelling caused by various stages of inflammation or infection of the pulpal or periradicular tissues. The cause of dental pain originates typically from caries, deep or defective restorations, or traumatic injuries. Sometimes occlusion-related issues such as cracks can also mimic acute dental pain (Fig. 19.1). Bender[9] stated that patients who manifest severe or referred pain almost always had a previous history of pain with the offending tooth. Approximately 85% of all dental emergencies arise as a result of pulpal or periapical disease, which would necessitate either extraction or endodontic treatment to relieve the symptoms.[43,75] It has also been estimated that about 12% of the United States population experienced a toothache in the preceding 6 months.[71]

Determining a definitive diagnosis can sometimes be challenging and even frustrating for clinicians, but a methodical, objective, and subjective evaluation is imperative before developing a proper treatment plan. Unfortunately, on the basis of the diagnosis, there are conflicting opinions on how to best clinically manage various endodontic emergencies. Surveys obtained from board-certified endodontists in 1977,[25,26] 1990,[34] and 2009[69] proposed to define seven conditions as endodontic emergencies. However, due to terminology changes, five clinical presentations are presently considered legitimate emergencies:

1. Symptomatic irreversible pulpitis with normal apical tissues
2. Symptomatic irreversible pulpitis with symptomatic apical periodontitis
3. Necrotic pulp with symptomatic apical periodontitis
4. Necrotic pulp with fluctuant intraoral swelling
5. Necrotic pulp with diffuse facial swelling

There are other endodontic emergencies such as traumatic dental injuries, previously endodontic treated teeth with either symptomatic apical periodontitis or acute apical abscess, and endodontic flare-ups that may occur between treatment sessions. Of course, there are also many types of facial pain that have a nonodontogenic origin; these are described in detail in Chapter 4.

There have been several changes in the determined clinical management of endodontic emergencies over the years. Many of these treatment modifications have occurred because of the more contemporary armamentarium and materials as well as new evidence-based research and the presumption of empirical clinical success.

Emergency Endodontic Management

Because pain is both a psychological and biologic entity, the management of acute dental pain must take into consideration both the physical symptoms and the emotional status of the patient. The patient's needs, fears, and coping mechanisms must be compassionately understood. This assessment and the clinician's ability to build rapport with the patient are critical factors in the comprehensive success of the patient's management.[9,33,54,99]

The methodical steps for determining an accurate diagnosis, based on evaluation of the patient's chief complaint, review of the medical history, and the protocols used for an objective and subjective diagnosis, are described in detail in

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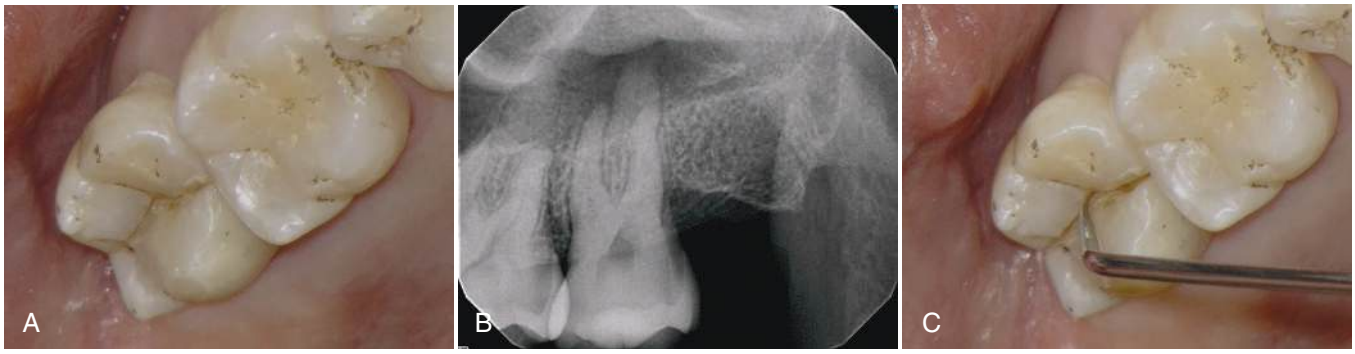


Fig. 19.1 **A,** Patient complained of acute pain on biting at the upper left molar (tooth #15). **B,** Clinical thermal tests revealed pulpal necrosis and a preoperative radiograph taken (periapical lesions noted). **C,** The tooth was split in half.

Chapter 1. Once it has been determined that endodontic treatment is necessary, it is incumbent on the clinician to take the proper steps necessary to manage the acute dental emergency.

The clinician has a responsibility to inform the patient of the recommended treatment plan and to advise the patient of the treatment alternatives, the risks, and benefits that pertain, and the expected prognosis under the present circumstances. Given this information, the patient may elect extraction over endodontics or possibly request a second opinion. The treatment plan should never be forced on a patient. The informed course of treatment is made jointly between the patient and the clinician.

In the event of an endodontic emergency, the clinician must determine the optimal mode of treatment according to the diagnosis. Treatment may vary depending on the pulpal or periapical status, the intensity and duration of pain, and whether there is diffuse or fluctuant swelling. Paradoxically, as discussed later, the mode of therapy that we tend to choose has been directed more from surveys of practicing endodontists rather than from controlled clinical studies or research investigations.

TEETH WITH VITAL PULPS

Teeth with vital pulps can have one of the following presentations:

- *Normal.* The teeth are asymptomatic with no objective pathoses.
- *Reversible pulpitis.* There is a reversible sensitivity to cold or osmotic changes (i.e., sweet, salty, and sour).
- *Irreversible pulpitis.* The sensitivity to temperature changes is more intense and with a longer duration.

REVERSIBLE PULPITIS

Reversible pulpitis can be induced by caries, exposed dentin, recent dental treatment, and defective restorations. Conservative removal of caries, protection of dentin, and a proper restoration will typically resolve the symptoms. However, the symptoms from exposed dentin, specifically from gingival recession and cervically exposed roots, can often be difficult to alleviate. Topical applications of desensitizing agents and the use of certain dentifrices have been helpful in the management of dentin hypersensitivity; the etiology, physiology, and management of this are discussed in Chapter 13.

IRREVERSIBLE PULPITIS

The diagnosis of irreversible pulpitis can be subcategorized as asymptomatic or symptomatic. *Asymptomatic* irreversible pulpitis pertains to a tooth that has no symptoms, but has deep caries or tooth structure loss that, if left untreated, will cause the tooth to become symptomatic or nonvital. On the other hand, the pain from *symptomatic* irreversible pulpitis is often an emergency condition that requires immediate treatment. These teeth exhibit intermittent or spontaneous pain and are sensitive to thermal triggers. Exposure to extreme temperatures, especially cold, will elicit intense and prolonged episodes of pain, even after the source of the stimulus is removed.

In 1977,[25,26] 187 board-certified endodontists responded to a survey to determine how they would manage various endodontic emergencies. Ten years later, 314 board-certified endodontists responded to the same questionnaire in order to determine whether there had been any changes in how these emergencies were managed.[34] The clinical management of emergency treatment of a tooth with symptomatic irreversible pulpitis with or without normal periapical tissues seemed to be reasonably similar by removing the inflamed pulp tissue either by pulpotomy or complete instrumentation.[83] In a similar survey conducted in 2009,[69] most respondents stated that they would clean to the level of the "apex," as confirmed with an electronic apex locator; this suggests a change in the management of endodontic cases based on the advent of a more contemporary armamentarium. In general, there is a trend toward more cleaning and shaping of the canal when symptomatic irreversible pulpitis presents with a normal periapex, compared with performing just pulpectomies as described in the 1977 survey. None of the individuals surveyed in the 1990 or 2009 poll stated that they would manage these emergencies by establishing any type of drainage by trephining the apex, making an incision, or leaving the tooth open for an extended period of time.

In addition, for vital teeth, the 1977 survey did not even broach the concept of completing the endodontics in one visit, whereas in the 1988 study about one third of the respondents indicated that they would complete these vital cases in a single visit and the response rose to 79% in the most recent survey. Since the early 1980s, there seems to have been an increase in the acceptability of providing endodontic therapy in one visit, especially in cases of vital pulps, with most studies revealing an equal number, or fewer, flare-ups after single-visit endodontic treatment.[27,86,91,96,98,106] However, this has not come without controversy, with some studies

showing otherwise,[118] contending that there is more posttreatment pain after single-visit endodontics, and possibly a lower long-term success rate. Unfortunately, time constraints at the emergency visit often make the single-visit treatment option not practical.[4]

If root canal therapy is completed at a later date, medicating the canal with calcium hydroxide has been suggested to reduce the chances of bacterial growth in the canal between appointments in most studies.[20] One randomized clinical study showed that a dry cotton pellet was as effective in relieving pain as a pellet moistened with camphorated monochlorophenol (CMCP), metacresylacetate (cresatin), eugenol, or saline.[45] Sources of infection, such as caries and defective restorations, should be completely removed to prevent recontamination of the root canal system between appointments.[45] The concept of single- versus multiple-visit endodontics is described in greater detail in Chapter 17.

For emergency management of vital teeth that are not initially sensitive to percussion, occlusal reduction has not been shown to be beneficial.[22,34] However, the clinician should be cognizant of the possibility of occlusal interferences and prematurities that might cause tooth fracture under heavy mastication. In vital teeth in which the inflammation has extended periapically, which will present with pretreatment pain to percussion, occlusal reduction has been reported to reduce posttreatment pain.[34,82,97]

Antibiotics are not recommended for the emergency management of irreversible pulpitis (see Chapters 15 and 17),[59,107] as placebo-controlled clinical trials have demonstrated that antibiotics have no effect on pain levels in patients with irreversible pulpitis.[80]

Most endodontists and endodontic textbooks recommend the emergency management of *symptomatic* irreversible pulpitis to involve the initiation of root canal treatment,[20,34,44,69,111] with complete pulp removal and total debridement of the root canal system. Unfortunately, in an emergency situation, the allotted time necessary for this treatment is often an issue. Given the potential time constraints and inevitable differences in skill level between clinicians, it may not be feasible to complete the total canal cleaning at the initial emergency visit. Subsequently, especially with multirooted teeth, a pulpotomy (removal of the coronal pulp) has been advocated for emergency treatment of irreversible pulpitis.[17,44,103] In a clinical study of various emergency procedures, it has been demonstrated that this treatment is highly effective for alleviating acute dental pain due to irreversible pulpitis.[17] A randomized clinical trial showed that pulpotomy was an excellent alternative for emergency relief of pain.[32] These findings were similar to the results seen in a recent study,[5] which reported more pain relief after pulpotomy as compared to the root canals treatment.

To assist the clinician in assessing the level of difficulty of a given endodontic case, the American Association of Endodontists (Chicago, IL) has developed the "AAE Endodontic Case Difficulty Assessment Form and Guidelines" (Fig. 19.2). This form is intended to make case selection more efficient, more consistent, and easier to document, as well as to provide a more objective ability to determine when it may be necessary to refer the patient to another clinician who may be better able to manage the complexities of the case. Haug et al. evaluated the impact of the case difficulty on endodontic mishaps in an undergraduate student clinic. They emphasized the importance and relevance of case difficulty assessment. Their results showed that the American Association of Endodontists (AAE) case difficulty assessment form is a valuable tool to predict potential mishaps and the number of treatment visits.[46]

PULPAL NECROSIS WITH SYMPTOMATIC APICAL PERIODONTITIS

Over the years, the proper methodology for the emergency endodontic management of necrotic teeth has been controversial. In a 1977 survey of board-certified endodontists,[25,26] it was reported that, in the absence of swelling, most respondents would completely instrument the canals, keeping the file short of the radiographic apex. However, when swelling was present, the majority of those polled in 1977 preferred to leave the tooth open, with instrumentation extending beyond the apex to help facilitate drainage through the canals. Years later and again validated in a 2009 study, most respondents favored complete instrumentation regardless of the presence of swelling. Also, it was the decision of 25.2% to 38.5% of the clinicians to leave these teeth open in the event of diffuse swelling; 17.5% to 31.5% left the teeth open in the presence of a fluctuant swelling. However, as discussed later, there is currently a trend toward not leaving teeth open for drainage. There is also another trend: when treatment is done in more than one visit, most endodontists will use calcium hydroxide as an intracanal medicament.[69]

Care should be taken not to push necrotic debris beyond the apex during root canal instrumentation, as this has been shown to promote more posttreatment discomfort.[15,34,95,104] Crown-down instrumentation techniques have been shown to remove most of the debris coronally rather than pushing it out the apex. The use of positive-pressure irrigation methods, such as needle-and-syringe irrigation, also poses a risk of expressing debris or solution out of the apex.[12,23] Improvements in technology, such as electronic apex locators and use of the limited cone-beam computed tomography (CBCT) have facilitated increased accuracy in determining working length measurements.[74] An increased number of clinicians now use these devices.[62,69] Moreover, new negative-pressure irrigation systems may allow for a more thorough canal debridement with less apical extrusion of debris.[18,75]

The attenuation of postoperative pain in patients with this specific condition has continuously been evaluated in the literature. An outcome study indicated that the use of a negative apical pressure irrigation device significantly reduces the postoperative pain levels in comparison to conventional needle irrigation.[3] A randomized clinical trial demonstrated final irrigation with 20 mL sterile cold (2.5°C) saline solution delivered to the working length with a sterile, cold (2.5°C) EndoVac microcannula (Kerr Endo, Orange Country, CA) for 5 minutes.[116] They concluded that cryotherapy reduced the incidence of postoperative pain and the need for medication in patients presenting with a diagnosis of necrotic pulp and symptomatic apical periodontitis.

Trephination

In the absence of swelling, trephination is the surgical perforation of the alveolar cortical plate to release, from between the cortical plates, the accumulated inflammatory and infective tissue exudate that causes pain. Its use has

been historically advocated to provide pain relief in patients with severe and recalcitrant periradicular pain.[25,26] The technique involves an engine-driven perforator entering through the cortical bone and into the cancellous bone, often without the need for an incision,[19] in order to provide a pathway for drainage from the periradicular tissues. Although more recent studies have failed to show the benefit of trephination in patients with irreversible pulpitis with symptomatic apical periodontitis[77] or necrotic teeth with symptomatic apical periodontitis,[82] there remain some advocates who recommend trephination for managing acute and intractable periapical pain.[51] The clinician should understand that local anesthesia may be difficult for cases with acute inflammation or infection.[55] Extreme care must be taken when carrying out a trephination procedure to guard against inadvertent and possibly irreversible injury to the tooth root or surrounding structures, such as the mental foramen, intra-alveolar nerve, or maxillary sinus.

Necrosis and Single-Visit Endodontics

Although single-visit endodontic treatment for teeth diagnosed with irreversible pulpitis is not contraindicated,[2,91,93,98,119] performing single-visit endodontics on necrotic and previously treated teeth is not without controversy. In cases

AAE Endodontic Case Difficulty Assessment Form and Guidelines

PATIENT INFORMATION

Name_____

Address_____

City/State/Zip_____

Phone_____

DISPOSITION

Treat in Office: Yes ☐ No ☐

Refer Patient to:

_____

Date:_____

Guidelines for Using the AAE Endodontic Case Difficulty Assessment Form

The AAE designed the Endodontic Case Difficulty Assessment Form for use in endodontic curricula. The Assessment Form makes case selection more efficient, more consistent and easier to document. Dentists may also choose to use the Assessment Form to help with referral decision making and record keeping.

Conditions listed in this form should be considered potential risk factors that may complicate treatment and adversely affect the outcome. Levels of difficulty are sets of conditions that may not be controllable by the dentist. Risk factors can influence the ability to provide care at a consistently predictable level and impact the appropriate provision of care and quality assurance.

The Assessment Form enables a practitioner to assign a level of difficulty to a particular case.

LEVELS OF DIFFICULTY

MINIMAL DIFFICULTY	Preoperative condition indicates routine complexity (uncomplicated). These types of cases would exhibit only those factors listed in the MINIMAL DIFFICULTY category. Achieving a predictable treatment outcome should be attainable by a competent practitioner with limited experience.
MODERATE DIFFICULTY	Preoperative condition is complicated, exhibiting one or more patient or treatment factors listed in the MODERATE DIFFICULTY category. Achieving a predictable treatment outcome will be challenging for a competent, experienced practitioner.
HIGH DIFFICULTY	Preoperative condition is exceptionally complicated, exhibiting several factors listed in the MODERATE DIFFICULTY category or at least one in the HIGH DIFFICULTY category. Achieving a predictable treatment outcome will be challenging for even the most experienced practitioner with an extensive history of favorable outcomes.

Review your assessment of each case to determine the level of difficulty. If the level of difficulty exceeds your experience and comfort, you might consider referral to an endodontist.

The AAE Endodontic Case Difficulty Assessment Form is designed to aid the practitioner in determining appropriate case disposition. The American Association of Endodontists neither expressly nor implicitly warrants any positive results associated with the use of this form. This form may be reproduced but may not be amended or altered in any way.

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Fig. 19.2 The American Association of Endodontists *(AAE)* Endodontic Case Difficulty Assessment Form and Guidelines, developed to assist the clinician in assessing the level of difficulty of a given endodontic case and to help determine when referral may be necessary.

AAE Endodontic Case Difficulty Assessment Form

Criteria and Subcriteria	Minimal Difficulty	Moderate Difficulty	High Difficulty
A. PATIENT CONSIDERATIONS			
Medical history	☐ No medical problem (ASA Class 1*)	☐ One or more medical problems (ASA Class 2*)	☐ Complex medical history/serious illness/disability (ASA Classes 3-5*)
Anesthesia	☐ No history of anesthesia problems	☐ Vasoconstrictor intolerance	☐ Difficulty achieving anesthesia
Patient disposition	☐ Cooperative and compliant	☐ Anxious but cooperative	☐ Uncooperative
Ability to open mouth	☐ No limitation	☐ Slight limitation in opening	☐ Significant limitation in opening
Gag reflex	☐ None	☐ Gags occasionally with radiographs/treatment	☐ Extreme gag reflex which has compromised past dental care
Emergency condition	☐ Minimum pain or swelling	☐ Moderate pain or swelling	☐ Severe pain or swelling
B. DIAGNOSTIC AND TREATMENT CONSIDERATIONS			
Diagnosis	☐ Signs and symptoms consistent with recognized pulpal and periapical conditions	☐ Extensive differential diagnosis of usual signs and symptoms required	☐ Confusing and complex signs and symptoms: difficult diagnosis ☐ History of chronic oral/facial pain
Radiographic difficulties	☐ Minimal difficulty obtaining/interpreting radiographs	☐ Moderate difficulty obtaining/interpreting radiographs (*e.g.*, high floor of mouth, narrow or low palatal vault, presence of tori)	☐ Extreme difficulty obtaining/interpreting radiographs (*e.g.*, superimposed anatomical structures)
Position in the arch	☐ Anterior/premolar ☐ Slight inclination (<10°) ☐ Slight rotation (<10°)	☐ 1st molar ☐ Moderate inclination (10-30°) ☐ Moderate rotation (10-30°)	☐ 2nd or 3rd molar ☐ Extreme inclination (>30°) ☐ Extreme rotation (>30°)
Tooth isolation	☐ Routine rubber dam placement	☐ Simple pretreatment modification required for rubber dam isolation	☐ Extensive pretreatment modification required for rubber dam isolation
Morphologic aberrations of crown	☐ Normal original crown morphology	☐ Full coverage restoration ☐ Porcelain restoration ☐ Bridge abutment ☐ Moderate deviation from normal tooth/root form (*e.g.*, taurodontism, microdens) ☐ Teeth with extensive coronal destruction	☐ Restoration does not reflect original anatomy/alignment ☐ Significant deviation from normal tooth/root form (*e.g.*, fusion, dens in dente)
Canal and root morphology	☐ Slight or no curvature (<10°) ☐ Closed apex <1 mm diameter	☐ Moderate curvature (10-30°) ☐ Crown axis differs moderately from root axis. Apical opening 1-1.5 mm in diameter	☐ Extreme curvature (>30°) or S-shaped curve ☐ Mandibular premolar or anterior with 2 roots ☐ Maxillary premolar with 3 roots ☐ Canal divides in the middle or apical third ☐ Very long tooth (>25 mm) ☐ Open apex (>1.5 mm in diameter)
Radiographic appearance of canal(s)	☐ Canal(s) visible and not reduced in size	☐ Canal(s) and chamber visible but reduced in size ☐ Pulp stones	☐ Indistinct canal path ☐ Canal(s) not visible
Resorption	☐ No resorption evident	☐ Minimal apical resorption	☐ Extensive apical resorption ☐ Internal resorption ☐ External resorption
C. ADDITIONAL CONSIDERATIONS			
Trauma history	☐ Uncomplicated crown fracture of mature or immature teeth	☐ Complicated crown fracture of mature teeth ☐ Subluxation	☐ Complicated crown fracture of immature teeth ☐ Horizontal root fracture ☐ Alveolar fracture ☐ Intrusive, extrusive or lateral luxation ☐ Avulsion
Endodontic treatment history	☐ No previous treatment	☐ Previous access without complications	☐ Previous access with complications (*e.g.*, perforation, non-negotiated canal, ledge, separated instrument) ☐ Previous surgical or nonsurgical endodontic treatment completed
Periodontal-endodontic condition	☐ None or mild periodontal disease	☐ Concurrent moderate periodontal disease	☐ Concurrent severe periodontal disease ☐ Cracked teeth with periodontal complications ☐ Combined endodontic/periodontic lesion ☐ Root amputation prior to endodontic treatment

*American Society of Anesthesiologists (ASA) Classification System

Class 1: No systemic illness. Patient healthy.
Class 2: Patient with mild degree of systemic illness, but without functional restrictions, *e.g.*, well-controlled hypertension.
Class 3: Patient with severe degree of systemic illness which limits activities, but does not immobilize the patient.
Class 4: Patient with severe systemic illness that immobilizes and is sometimes life threatening.
Class 5: Patient will not survive more than 24 hours whether or not surgical intervention takes place.

www.asahq.org/clinical/physicalstatus.htm

Fig. 19.2, cont'd

of necrotic teeth, although research[27] has indicated that there may be no difference in posttreatment pain if the canals are filled at the time of the emergency versus a later date, some studies[105,112] have questioned the long-term prognosis of such treatment, especially in cases of symptomatic apical periodontitis. Several studies,[28,66] including a CONSORT (Consolidated Standards of Reporting Trials) meta-analysis,[92] have shown no difference in outcome between single-visit and two-visit treatments. The concept of single- versus multi-visit endodontics is further discussed in Chapters 5 and 17.

PULPAL NECROSIS WITH ACUTE APICAL ABSCESS

Swelling

Tissue swelling may be associated with an acute apical abscess at the time of the initial emergency visit, or it may occur as an inter-appointment flare-up or as a postendodontic complication. Swellings may be localized or diffuse, fluctuant or firm. Localized swellings are confined within the oral cavity, whereas a diffuse swelling, or cellulitis, is more extensive, spreading through adjacent soft tissues and dissecting tissue spaces along fascial planes.[100]

Swelling may be controlled by establishing drainage through the root canal or by incising the fluctuant swelling. As discussed later and in Chapter 15, antibiotics may be recruited when there are systemic manifestations of the infection, such as fever and malaise. The principal modality for managing swelling secondary to endodontic infections is to achieve drainage and remove the source of the infection.[41] When the swelling is localized, the preferred avenue is drainage through the root canal (Fig. 19.3). Complete canal debridement and disinfection[42,114] are paramount for success regardless of observable drainage, because the presence of any bacteria remaining within the root canal system will compromise the resolution of the acute infection.[73] In the presence of persistent swelling, gentle finger pressure to the mucosa overlying the swelling may help facilitate drainage through the canal. Once the canals have been cleaned and allowed to dry, calcium hydroxide as the intracanal medicament[69] should be placed and the access properly sealed.[20,34,44]

Fascial Space Infections

If bacteria from the infected root canal gain entry into the periradicular tissues and the immune system is unable to suppress the invasion, an otherwise healthy patient eventually shows signs and symptoms of an acute apical abscess, which can, in turn, evolve to cellulitis. Clinically, the patient experiences swelling and mild to severe pain. Depending on the relationship of the apices of the affected tooth to the muscular attachments, the swelling may be localized to the

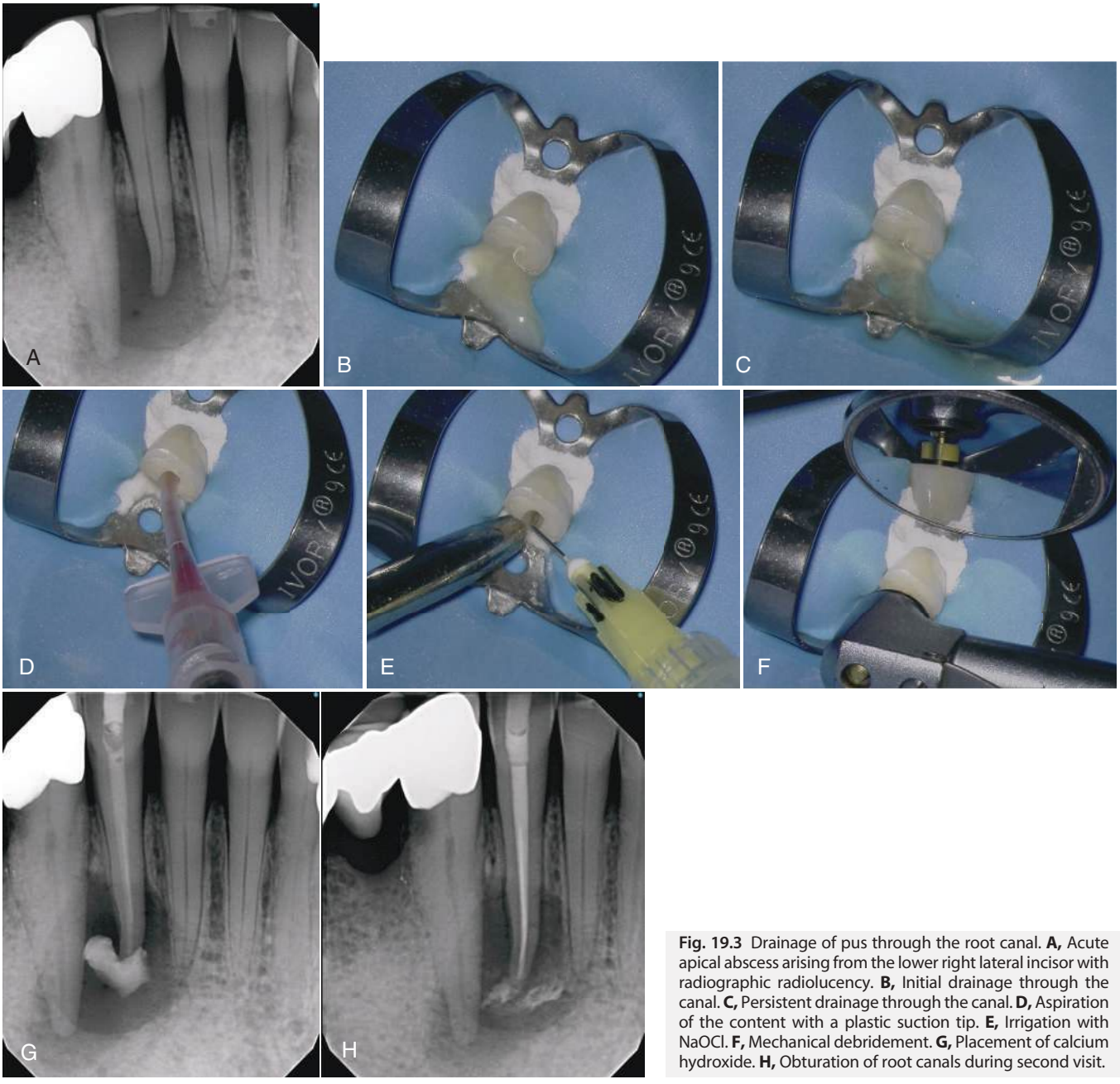


Fig. 19.3 Drainage of pus through the root canal. **A,** Acute apical abscess arising from the lower right lateral incisor with radiographic radiolucency. **B,** Initial drainage through the canal. **C,** Persistent drainage through the canal. **D,** Aspiration of the content with a plastic suction tip. **E,** Irrigation with NaOCl. **F,** Mechanical debridement. **G,** Placement of calcium hydroxide. **H,** Obturation of root canals during second visit.

vestibule or extend into a fascial space. The patient may also have systemic manifestations, such as fever, chills, lymphadenopathy, headache, and nausea. Because the reaction to the infection may occur quickly, the involved tooth may or may not show radiographic evidence of a widened periodontal ligament space. In most cases, the tooth elicits a positive response to percussion, and the periradicular area is tender to palpation. The tooth now becomes a *focus of infection* because it leads to periradicular infection and secondary spread to the fascial spaces of the head and neck, resulting in cellulitis and systemic signs and symptoms of infection.

In such cases, treatment may involve incision for drainage, root canal treatment, or extraction to remove the source of the infection. Antibiotic therapy may be indicated in patients with compromised host resistance, the presence of systemic symptoms, or fascial space involvement. Fascial space infections of odontogenic origin are infections that have spread into the fascial spaces from the periradicular area of a tooth, the focus of infection. They are *not* examples of the theory of focal infection, which describes the dissemination of bacteria or their products from a distant focus of infection. Preferably, this is an example of the local spread of infection from an odontogenic source.

Fascial spaces are potential anatomic areas that exist between the fascia and underlying organs and other tissues. During infection, these spaces are formed as a result of the spread of purulent exudate. The spread of infections of odontogenic origin into the fascial spaces of the head and neck is determined by the location of the root end of affected tooth to its overlying buccal or lingual cortical plate and the relationship of the apex to the attachment of a muscle (Fig. 19.4, *A*). For example, if the source of the infection is a mandibular molar whose apices lie closer to the lingual cortical plate and *above* the attachment of the mylohyoid muscle of the floor of the mouth, the purulent exudate may break through the lingual cortical plate and into the *sublingual* space. However, if the apices lie *below* the attachment of the mylohyoid muscle, the infection may spread into the *submandibular* space.

As described by Hohl and colleagues,[53] the fascial spaces of the head and neck can be categorized into four anatomic groups:

- The mandible and below
- The cheek and lateral face
- The pharyngeal and cervical areas
- The midface

Swellings of and below the mandible include six anatomic areas or fascial spaces:

- The buccal vestibule
- The body of the mandible
- The mental space
- The submental space
- The sublingual space
- The submandibular space

The *mandibular buccal vestibule* is the anatomic area amid the buccal cortical plate, the overlying alveolar mucosa, and the buccinator muscle (posterior) or the mentalis muscle (anterior) (see Fig. 19.4, *B* and *C*). In this case, the source of the infection is a mandibular posterior or anterior tooth in which the purulent exudate breaks through the buccal cortical plate, and the apex or apices of the involved tooth lie above the attachment of the buccinator or mentalis muscle, respectively.

The *space of the body of the mandible* is the potential anatomic area between the buccal or lingual cortical plate and its overlying periosteum. The source of infection is a mandibular tooth in which the purulent exudate has broken through the overlying cortical plate but not yet perforated the overlying periosteum. Involvement of this space can also occur as a result of a postsurgical infection.

The *mental space* (see Fig. 19.4, *D*) is the potential bilateral anatomic area of the chin that lies between the mentalis muscle superiorly and the platysma muscle inferiorly. The source of the infection is an anterior tooth in which the purulent exudate breaks through the buccal cortical plate, and the apex of the tooth lies below the attachment of the mentalis muscle.

The *submental space* (see Fig. 19.4, *E*) is the potential anatomic area between the mylohyoid muscle superiorly and the platysma muscle inferiorly. The source of the infection is an anterior tooth in which the purulent exudate breaks through the lingual cortical plate, and the apex of the tooth lies below the attachment of the mylohyoid muscle.

The *sublingual space* (see Fig. 19.4, *F*) is the potential anatomic area between the oral mucosa of the floor of the mouth superiorly and the mylohyoid muscle inferiorly. The lateral boundaries of the space are the lingual surfaces of the mandible. The source of infection is any mandibular tooth in which the purulent exudate breaks through the lingual cortical plate, and the apex or apices of the tooth lie above the attachment of the mylohyoid muscle.

The *submandibular space* (see Fig. 19.4, *G*) is the potential space between the mylohyoid muscle superiorly and the platysma muscle inferiorly. The source of infection is a posterior tooth, usually a molar, in which the purulent exudate breaks through the lingual cortical plate and the apices of the tooth lie below the attachment of the mylohyoid muscle. If the submental, sublingual, and submandibular spaces are involved at the same time, a diagnosis of Ludwig angina is made. This life-threatening cellulitis can advance into the pharyngeal and cervical spaces, resulting in airway obstruction.

Swellings of the lateral face and cheek include four anatomic areas or fascial spaces:

- The buccal vestibule of the maxilla
- The buccal space
- The submasseteric space
- The temporal space

Anatomically, the *buccal vestibular space* (see Fig. 19.4, *H*) is the area between the buccal cortical plate, the overlying mucosa, and the buccinator muscle. The superior extent of the space is the attachment of the buccinator muscle to the zygomatic process. The source of infection is a posterior maxillary tooth in which the purulent exudate breaks through the buccal cortical plate, and the apex of the tooth lies below the attachment of the buccinator muscle.

The *buccal space* (see Fig. 19.4, *I*) is the potential space between the lateral surface of the buccinator muscle and the medial surface of the skin of the cheek. The superior extent of the space is the attachment of the buccinator muscle to the zygomatic arch, whereas the inferior and posterior boundaries are the attachment of the buccinator to the inferior border of the mandible and the anterior margin

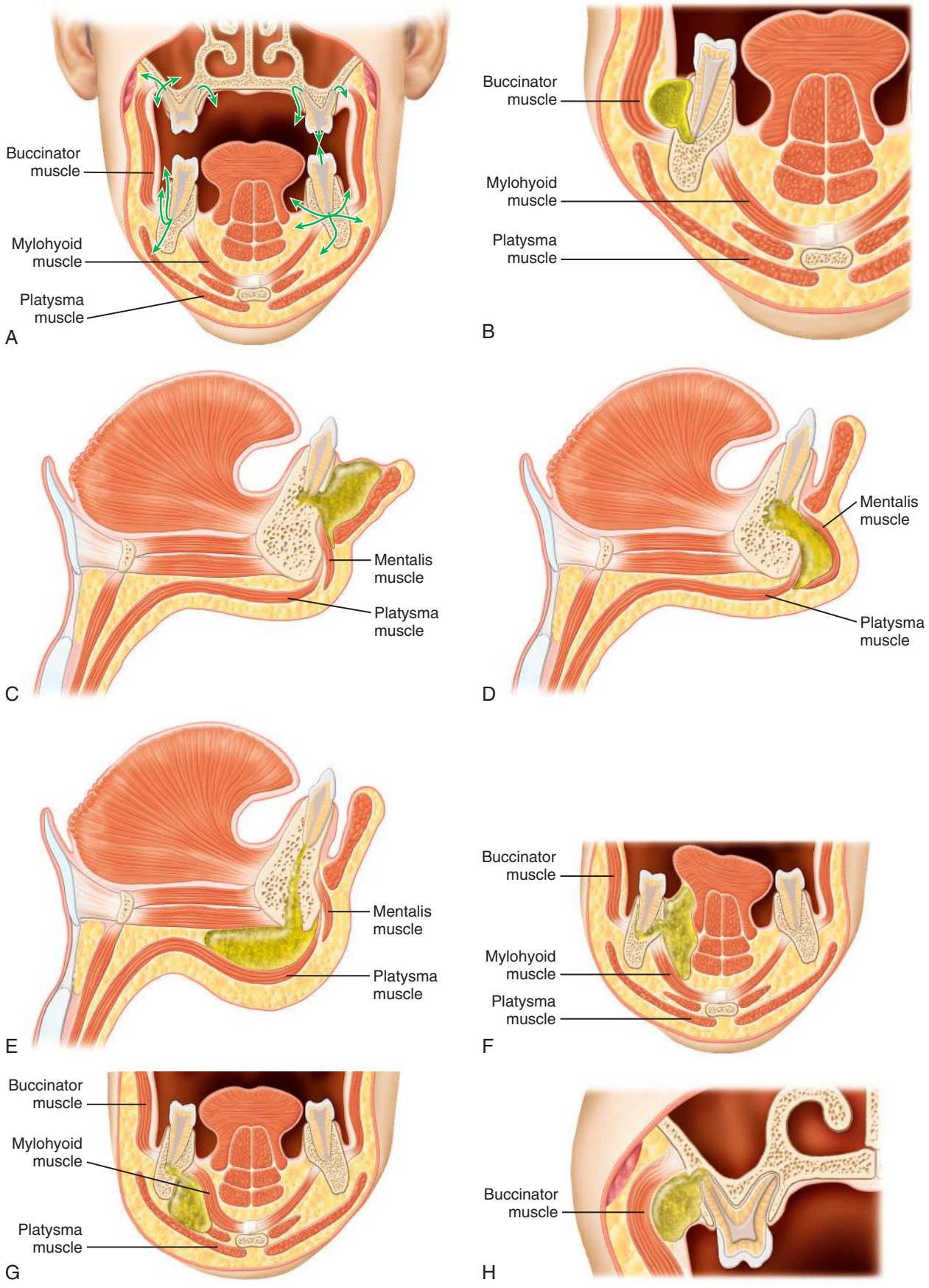
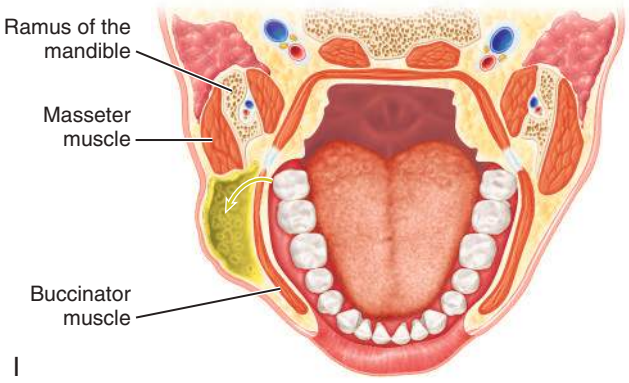
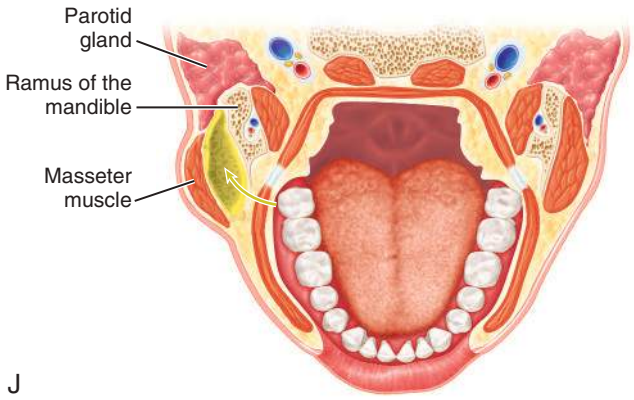


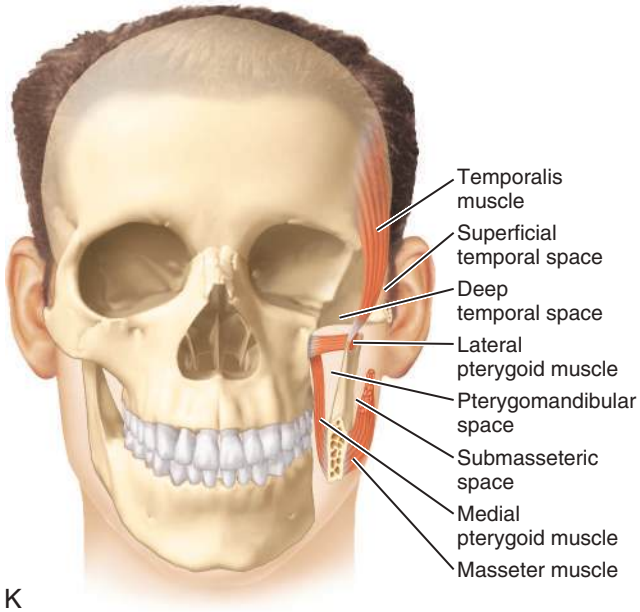
Fig. 19.4 A, Spread of odontogenic infections. **B,** Mandibular buccal vestibule (posterior tooth). **C,** Mandibular buccal vestibule (anterior tooth). **D,** Mental space. **E,** Submental space. **F,** Sublingual space. **G,** Submandibular space. **H,** Maxillary buccal vestibule. **I,** Buccal space. **J,** Submasseteric space. **K,** Temporal space. **L,** Pterygomandibular space. **M,** Parapharyngeal spaces. **N,** Cervical spaces. **O,** Canine (infraorbital) and periorbital space.



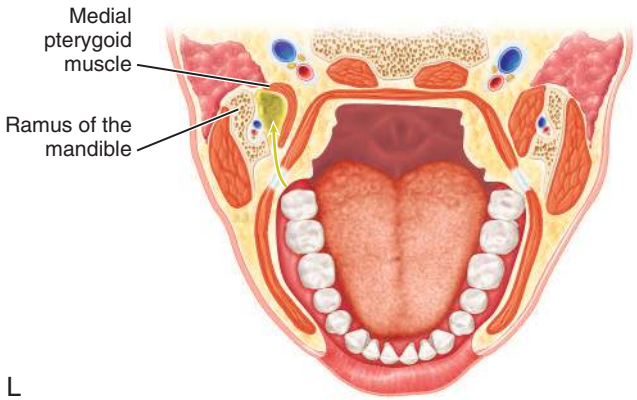
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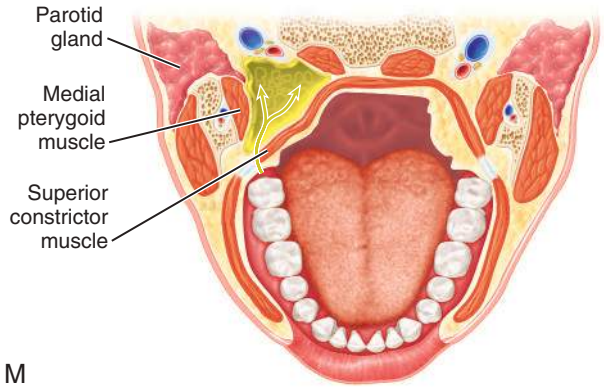
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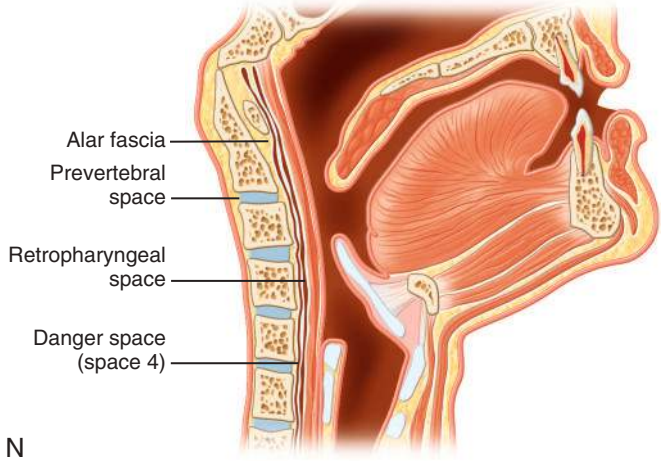
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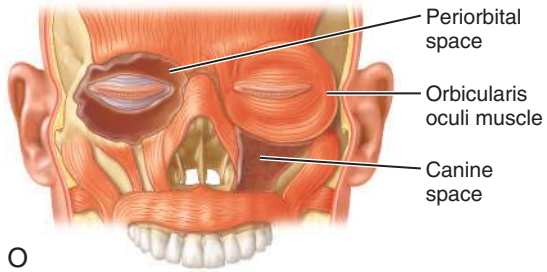
L



M



N



O

Fig. 19.4, cont'd

of the masseter muscle, respectively. The source of the infection can be either a posterior mandibular or maxillary tooth in which the purulent exudate breaks through the buccal cortical plate, and the apex or apices of the tooth lie above the attachment of the buccinator muscle (i.e., maxilla) or below the attachment of the buccinator muscle (i.e., mandible).

As the name implies, the *submasseteric space* (see Fig. 19.4, *J*) is the potential space between the lateral surface of the ramus of the mandible and the medial surface of the masseter muscle. The source of the infection is usually an impacted third molar in which the purulent exudate breaks through the lingual cortical plate, and the apices of the tooth lie very close to or within the space.

The *temporal space* (see Fig. 19.4, *K*) is divided into two compartments by the temporalis muscle. The *deep temporal space* is the potential space between the lateral surface of the skull and the medial surface of the temporalis muscle; the *superficial temporal space* lies between the temporalis muscle and its overlying fascia. The deep or superficial temporal spaces are involved indirectly if an infection spreads superiorly from the inferior pterygomandibular or submasseteric spaces, respectively.

Swellings of the pharyngeal and cervical areas include the following fascial spaces:

- The pterygomandibular space
- The parapharyngeal spaces
- The cervical spaces

The *pterygomandibular space* (see Fig. 19.4, *L*) is the potential space between the lateral surface of the medial pterygoid muscle and the medial surface of the ramus of the mandible. The superior extent of the space is the lateral pterygoid muscle. The source of the infection is mandibular second or third molars in which the purulent exudate drains directly into the space. In addition, contaminated inferior alveolar nerve injections can lead to infection of the space.

The *parapharyngeal spaces* comprise the lateral pharyngeal and retropharyngeal spaces (see Fig. 19.4, *M*). The *lateral pharyngeal space* is bilateral and lies between the lateral surface of the medial pterygoid muscle and the posterior surface of the superior constrictor muscle. The superior and inferior margins of the space are the base of the skull and the hyoid bone, respectively, and the posterior margin is the *carotid space,* or sheath, which contains the common carotid artery, internal jugular vein, and the vagus nerve. Anatomically, the *retropharyngeal space* lies between the anterior surface of the prevertebral fascia and the posterior surface of the superior constrictor muscle and extends inferiorly into the retroesophageal space, which extends into the posterior compartment of the mediastinum. The pharyngeal spaces usually become involved as a result of the secondary spread of infection from other fascial spaces or directly from a peritonsillar abscess.

The cervical spaces comprise the pretracheal, retrovisceral, danger, and prevertebral spaces (see Fig. 19.4, *N*). The *pretracheal space* is the potential space surrounding the trachea. It extends from the thyroid cartilage inferiorly into the superior portion of the anterior compartment of the mediastinum to the level of the aortic arch. Because of its anatomic location, odontogenic infections do not spread to the pretracheal space. The *retrovisceral space* comprises the retropharyngeal space superiorly and the retroesophageal space inferiorly. The space extends from the base of the skull into the posterior compartment of the mediastinum to a level between vertebrae C6 and T4. The *danger space* (i.e., space 4)[29,38] is the potential space between the alar and prevertebral fascia. Because this space is composed of loose connective tissue, it is considered an actual anatomic space extending from the base of the skull into the posterior compartment of the mediastinum to a level corresponding to the diaphragm. It is not unknown for odontogenic infection to spread into this space, if left untreated and undiagnosed.[36] The consequence can be fatal. The *prevertebral space* is the potential space surrounding the vertebral column. As such, it extends from vertebra C1 to the coccyx. A retrospective study showed that 71% of cases in which the mediastinum was involved arose from the spread of infection from the retrovisceral space (21% from the carotid space and 8% from the pretracheal space).[70]

Swellings of the midface consist of four anatomic areas and spaces:

- The palate
- The base of the upper lip
- The canine spaces
- The periorbital spaces

Odontogenic infections can spread into the areas between the palate and its overlying periosteum and mucosa and the base of the upper lip, which lies superior to the orbicularis oris muscle, even though these areas are not considered actual fascial spaces. The source of infection of the palate is any of the maxillary teeth in which the apex of the involved tooth lies close to the palate. The source of infection of the base of the upper lip is a maxillary central incisor in which the apex lies close to the buccal cortical plate and above the attachment of the orbicularis oris muscle.

The *canine,* or *infraorbital, space* (see Fig. 19.4, *O*) is the potential space between the levator anguli oris muscle inferiorly and the levator labii superioris muscle superiorly. The source of infection is the maxillary canine or first premolar in which the purulent exudate breaks through the buccal cortical plate, and the apex of the tooth lies above the attachment of the levator anguli oris muscle. There is a chance for the infection to spread from the infraorbital space into the cavernous sinus in the cranium via the valveless veins of the face and anterior skull base.[29]

The *periorbital space* (see Fig. 19.4, *O*) is the potential space that lies deep to the orbicularis oculi muscle. This space becomes involved through the spread of infection from the canine or buccal spaces. Infections of the midface can be very dangerous because they can result in *cavernous sinus thrombosis,* a life-threatening infection in which a thrombus formed in the cavernous sinus breaks free, resulting in blockage of an artery or the spread of infection. Under normal conditions, the angular and ophthalmic veins and the pterygoid plexus of veins flow into the facial and external jugular veins. If an infection has spread into the midfacial area, however, edema and the resultant increased pressure from the inflammatory response cause the blood to back up into the cavernous sinus. Once in the sinus, the blood stagnates and clots. The resultant infected thrombi remain in the cavernous sinus or escape into the circulation.[85,122]

Management of Abscesses and Cellulitis

The two most essential elements of effective patient management for the resolution of an odontogenic infection are correct diagnosis and removal of the cause. When endodontic treatment is possible and preferred, in an otherwise healthy patient, chemomechanical preparation of the infected root canals and incision for drainage of any fluctuant periradicular swelling usually provide immediate improvement of the clinical signs and symptoms. The majority of cases of endodontic infections can be treated effectively without the use of systemic antibiotics. The appropriate treatment is the removal of the cause of the inflammatory condition.

Antibiotics are still prescribed in clinical situations for which they are typically not indicated.[35] The evidence clearly showed that they are *not* recommended for irreversible pulpitis, symptomatic apical periodontitis, draining sinus tracts, after endodontic surgery, to prevent flare-ups, or after incision for drainage of a localized swelling (without cellulitis, fever, or lymphadenopathy).[31,52,80,94,118] When the ratio of risk to benefit is considered in these situations, antibiotic use may put the patient at risk for side effects of the antimicrobial agent and select for resistant microorganisms. Analgesics (not antibiotics) are indicated for controlling the pain.

Antibiotics in conjunction with appropriate endodontic treatment are recommended for progressive or persistent infections with systemic signs and symptoms such as fever (over 100°F or 37°C), malaise, cellulitis, unexplained trismus, and progressive or persistent swelling (or both) (Fig. 19.5). In such cases, antibiotic therapy is indicated as an adjunct to debridement of the root canal system, which is a reservoir of microorganisms. In addition, aggressive incision for drainage is indicated for any infection marked by cellulitis. The incision for drainage is indicated whether the cellulitis is indurate or fluctuant. It is essential to provide a pathway of drainage to prevent further spread of the abscess or cellulitis. An incision for drainage allows decompression of the increased tissue pressure associated with edema and provides significant pain relief. Furthermore, the incision provides a draining pathway not only for bacteria and their products but also for the inflammatory mediators associated with the spread of cellulitis.

A minimum inhibitory concentration of antibiotic may not reach the source of the infection because of decreased blood flow and because the antibiotic must diffuse through the edematous fluid and pus. Drainage of edematous fluid and purulent exudate improves circulation to the tissues associated with an abscess or cellulitis, providing better delivery of the antibiotic to the area. Placement of a drain may not be indicated for localized fluctuant swellings if complete evacuation of the purulent exudate is believed to have occurred.

For effective drainage, a vertically oriented stab incision is made through the mucoperiosteum in the most dependent site of the swelling. The incision must be long enough to allow blunt dissection using a curved hemostat or periosteal elevator under the periosteum for drainage of pockets of inflammatory exudate (Fig 19.6). A rubber dam drain or a Penrose drain is indicated for any patient with a progressive abscess or cellulitis to maintain an open pathway for drainage. A more detailed description is given later.

Patients with cellulitis should be followed daily to ensure that the infection is resolving. The best practical guide for determining the duration of antibiotic therapy is the clinical improvement of the patient. When clinical evidence indicates that the infection is certain to resolve or is resolved, antibiotics should be administered for no longer than 1 or 2 days more.

Endodontic treatment should be completed as soon as possible after the incision for drainage. The drain can usually be removed 1 or 2 days after improvement. If no significant clinical improvement occurs, the diagnosis and treatment must be reviewed carefully. Consultation with a specialist and referral may be indicated for severe or persistent infections. Likewise, patients requiring extraoral drainage should be referred to a clinician trained in the technique.



Fig. 19.5 **A,** Preoperative radiograph. **B,** Cellulitis.

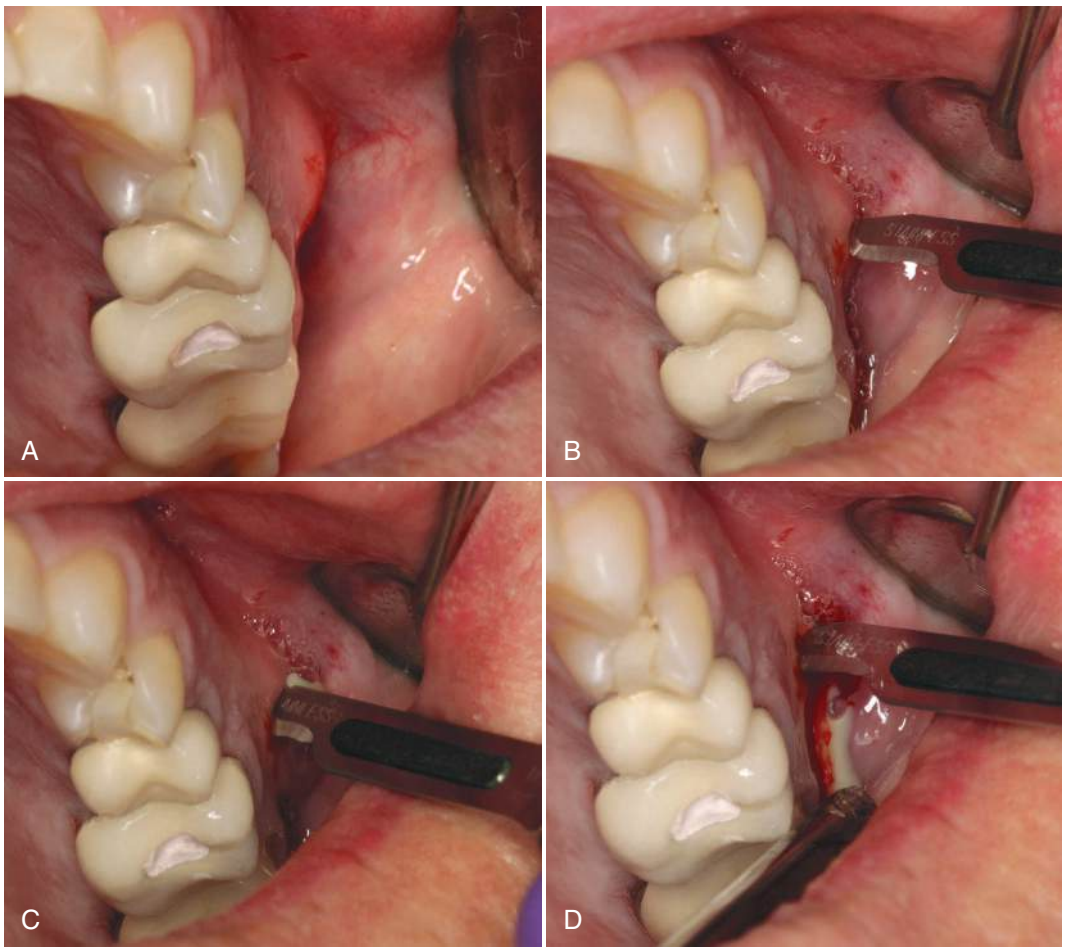


Fig. 19.6 **A,** Intraoral localized swelling. **B,** Preparation for incision. **C,** Incision. **D,** Purulence discharge.

Incision for Drainage

Although a recent study concluded that establishing drainage from a localized soft-tissue swelling may not result in the immediate relief of pain,[11] this procedure is sometimes necessary. This can be accomplished through the *incision for drainage* of the area.[81] Incision for drainage is indicated whether the cellulitis is indurated or fluctuant,[100] and it is used when a pathway for drainage is needed to prevent further spread of infection. The incision for drainage allows decompression from the increased tissue pressure associated with edema and can provide significant pain relief for the patient; noteworthy is that often pain decreases when the soft tissues swell due to the relief of pressure within the bone. The incision also provides a pathway not only for bacteria and bacterial by-products but also for the inflammatory mediators that are associated with the spread of cellulitis.

The basic principles of incision for drainage are as follows:

- Anesthetize the area.
- Make a vertically oriented incision at the site of greatest fluctuant swelling.
- Dissect gently, through the deeper tissues, and thoroughly explore all parts of the abscess cavity, eventually extending to the offending roots that are responsible for the pathosis. This will allow compartmentalized areas of inflammatory exudates and infection to be disrupted and evacuated.
- To promote drainage, the wound should be kept clean with warm saltwater mouth rinses. Intraoral heat application to infected tissues results in a dilation of small vessels, which subsequently intensifies host defenses through increased vascular flow.[42,100]
- A drain should be placed to prevent the incision from closing too early. One type of drain is ½-inch iodoform gauze, which is comfortable and less traumatic to the patient (Fig. 19.7). The patient should be seen the following day to remove the drain.

A diffuse swelling may develop into a life-threatening medical emergency. Because the spread of infection can traverse between the fascial planes and muscle attachments, vital structures can be compromised and breathing may be impeded. Two examples are Ludwig angina and cervical fasciitis.[36] It is imperative that the clinician is in constant communication with the patient to ensure that the infection does not worsen and that medical attention is provided as necessary. Antibiotics and analgesics should be prescribed, and the patient should be monitored closely for the next several days or until there is an improvement.

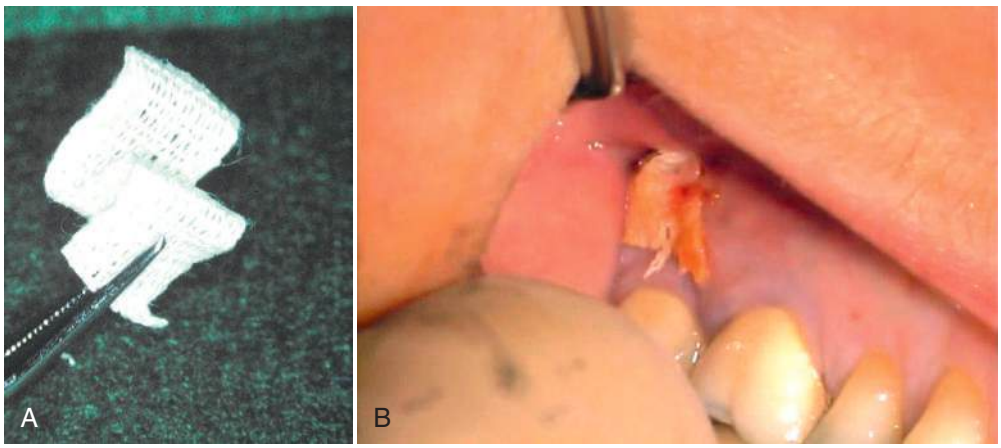


Fig. 19.7 **A,** Iodoform gauze cut to proper length. **B,** Iodoform gauze drain after 24 hours.

Individuals who show signs of toxicity, elevated body temperature, lethargy, central nervous system (CNS) changes, or airway compromise should be referred to an oral surgeon or medical facility for immediate care and intervention.

Symptomatic Teeth With Previous Endodontic Treatment

The emergency management of teeth with previous endodontic treatment may be technically challenging and time-consuming. This is especially true in the presence of extensive restorations, including posts and cores, crowns, and bridgework. However, the goal remains the same as for the management of necrotic teeth: remove contaminants from the root canal system and establish patency to achieve drainage. Gaining access to the periapical tissues through the root canals may require removal of posts and obturation materials, as well as negotiating blocked or ledged canals. Failure to complete root canal debridement and achieve periapical drainage may result in continued painful symptoms, in which case a trephination procedure or apicoectomy may be indicated. The ability, practicality, and feasibility to adequately retreat the root canal system must be carefully assessed before the initiation of treatment, as conventional retreatment might not be the optimal treatment plan. This is further discussed in Chapter 10.

Leaving Teeth Open

On rare occasions, canal drainage may continue from the periapical spaces (Fig. 19.8). In these cases, the clinician may opt to step away from the patient for some time to allow the drainage to continue and hopefully resolve on the same treatment visit.[111]

Historically, in the presence of acutely painful necrotic teeth with no swelling or diffuse swelling, 19.4% to 71.2% of surveyed endodontists would leave the tooth open between visits.[25,26] However, the more current literature makes it clear that this form of treatment would impair uneventful resolution and create a more complicated procedure.[6,8,120] For this reason, *leaving teeth open between appointments is not recommended*. Foreign objects have been found in teeth left open for drainage (Fig. 19.9). There has even been a documented case report of a foreign object being found to enter the periapical tissues through a tooth that had been left open for drainage.[103] In addition, leaving a tooth open provides an opportunity for oral microorganisms to invade and colonize the root canal system if the tooth is left open for an extended period.

Systemic Antibiotics for Endodontic Infections

A hundred years ago, infectious diseases were the major recognized causes of death in the world. The advent of antibiotics resulted in a significant decline in the incidence of life-threatening infections and heralded a new era in the therapy of infectious diseases, but the enthusiasm generated turned out to be premature. Over the years, microbial evolutionary responses to the selective pressure exerted by antibiotics have resulted in microbial species resistant to virtually every known antibiotic agent.[47] The rapid emergence of resistant microbial strains comes as a consequence of the astonishing power of natural selection among microorganisms. If a given member of a microbial community possesses genes of resistance against a certain antibiotic and the community is persistently exposed to the drug, the resistant microorganism is selected to emerge and multiply to the detriment of the susceptible portion of the community. Passing on the genes responsible for resistance via plasmids and quorum sensing[47] has also been shown to encourage the survival of the microbial community. The emergence of multidrug-resistant strains of several bacterial species capable of causing life-threatening infections has been reported.[47,72,90,108,121] Antibiotic resistance among obligately anaerobic bacteria is increasing, with resistance to penicillins, clindamycin, and cephalosporins noted at community hospitals and major medical centers.[48,49]

Among oral bacteria, there have been reports on emerging resistance to commonly used antibiotics. Resistance has been found in *Fusobacterium nucleatum* strains for penicillin, amoxicillin, and metronidazole, in *Prevotella intermedia* for tetracycline and amoxicillin, and in *Aggregatibacter*

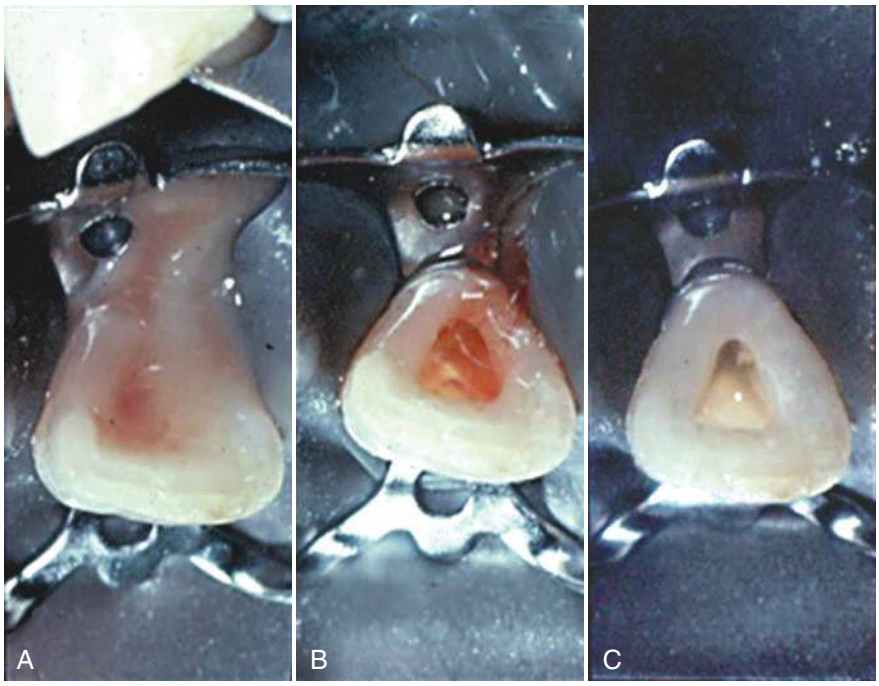


Fig. 19.8 Nonvital infected tooth with active drainage from the periapical area through the canal. **A,** Access opened and draining for 1 minute. **B,** Drainage after 2 minutes. **C,** Canal space dried after 3 minutes.



Fig. 19.9 Foreign object in tooth left open to drain. Patient used a sewing needle to clear out food particles that were blocking the canal and broke the needle in the tooth.

actinomycetemcomitans for amoxicillin and azithromycin.[64,115] Macrolides (erythromycin and azithromycin) have presented decreased activity against *Fusobacterium* and nonpigmented *Prevotella* species.[50,64,65] Production of beta-lactamase by oral bacteria has also been reported, with the most prominent beta-lactamase-producing bacteria belonging to the anaerobic genus *Prevotella*.[10,13,30,39,115] Kuriyama and colleagues[64] revealed that beta-lactamase production was detected in 36% of the black-pigmented *Prevotella* and 32% of the nonpigmented *Prevotella* species isolated from pus samples of oral abscesses. Susceptibility of *Prevotella* strains to several cephalosporins, erythromycin, and azithromycin has been found to correlate with amoxicillin susceptibility; amoxicillin-resistant strains can be similarly resistant to these other antibiotics.[64] This finding suggests that there is little value in the use of oral cephalosporins and macrolides in managing endodontic abscesses, mainly when penicillin-resistant strains are evident. Other enzyme-producing oral anaerobic species include strains of *F. nucleatum, Propionibacterium acnes, Actinomyces* species, and *Peptostreptococcus* species.[13,30,39,115] Facultative bacteria such as *Capnocytophaga* and *Neisseria* species have also been detected among the beta-lactamase producers.[39] Bacteria that produce beta-lactamases protect not only themselves but also other penicillin-susceptible bacteria present in a mixed community by releasing free beta-lactamase into the environment.[13]

Overuse and misuse of antibiotics have been considered the primary cause for the emergence of multidrug-resistant strains. Improper use of antibiotics includes use in cases with no infection, wrong choice of the agent, dosage or duration of therapy, and excessive consumption in prophylaxis.[88,89] Antibiotics are used in clinical practice far more often than necessary. Antibiotic therapy is warranted in about 20% of the individuals who are seen for clinical infectious disease, but they are prescribed up to 80% of the time. To complicate matters further, in up to 50% of cases, recommended agents, doses, or duration of therapy are incorrect.[69]

The appalling rise in the frequency of multidrug resistance among leading pathogens should cause great concern and incite a commitment to act carefully and responsibly. A single erroneous use of antibiotics can be a significant contribution to the current scenario of increasing microbial resistance. Diseases that were effectively treated in the past with a given antibiotic may now require the use of another drug, usually more expensive and potentially more toxic, to achieve effective antimicrobial treatment. Unfortunately, even the new prescription may not be valid.

Antibiotics are defined as naturally occurring substances of microbial origin or similar synthetic (or semisynthetic) elements that have antimicrobial activity in low concentrations and inhibit the growth of or kill selective microorganisms. The purpose of antibiotic therapy is to aid the host defenses in controlling and eliminating microorganisms that temporarily have overwhelmed the host defense mechanisms.[88] Based on the earlier discussion, it becomes clear that the most critical decision in antibiotic therapy is not so much *which* antibiotic should be employed but *whether* antibiotics should be used at all.[87] One should bear in mind that antibiotics are beneficial drugs classically employed to treat or help treat infectious disease and provide prophylaxis in carefully selected cases.

The majority of infections of endodontic origin are treated without the need for antibiotics. As mentioned, the absence of blood circulation in a necrotic pulp prevents antibiotics from reaching and eliminating microorganisms present in the root canal system; therefore, the source of infection is often unaffected by systemic antibiotic therapy. Antibiotics can, however, help impede the spread of the disease and development of focal infections in medically compromised patients and provide a valuable adjunct for managing selected cases of endodontic infection. In addition to the indications for systemic antibiotics discussed earlier for acute abscesses and cellulitis, antibiotics are also prescribed for prophylaxis in medically compromised patients during routine endodontic therapy, in some cases of persistent exudation not resolved after revision of intracanal procedures, and after the replantation of avulsed teeth.

Selection of antibiotics in clinical practice is either empirical or based on the results of microbial sensitivity tests. For diseases with known microbial causes, empirical therapy may be used. This is especially applicable to infections of endodontic origin because culture-dependent antimicrobial analyses of anaerobic bacteria can take too long to provide results about their susceptibility to antibiotics (7 to 14 days). Therefore, it is preferable to opt for an antimicrobial agent whose spectrum of action includes the most commonly detected bacteria. Most of the bacterial species involved with endodontic infections, including abscesses, are susceptible to penicillins,[7,57,61,65] which make them first-line drugs of choice. Because the use of antibiotics is restricted to severe infections or prophylaxis, it seems prudent to use amoxicillin, a semisynthetic penicillin with a broad spectrum of antimicrobial activity and one that is well absorbed in the alimentary canal. In more severe cases, including life-threatening conditions, combining amoxicillin with clavulanic acid or metronidazole may be required to achieve optimum antimicrobial effects as a result of the extended spectrum of action to include penicillin-resistant strains.[65] In patients allergic to penicillins or cases refractory to amoxicillin therapy, clindamycin is indicated. Clindamycin has potent antimicrobial activity against oral anaerobes.[61,63,65,67]

The risk/benefit ratio should always be evaluated before prescribing antibiotics. Appropriately selected patients will benefit from systemically administered antibiotics. Restrictive and conservative use of antibiotics is highly recommended in endodontic practice. Indiscriminate use (including cases of reversible or irreversible pulpitis) is contrary to sound clinical practice, as it may cause a selective overgrowth of intrinsically resistant bacteria, predisposing patients to secondary and superinfections, rendering drugs ineffective against potentially fatal medical infectious diseases.

Analgesics

As a more thorough description of pain medications can be found in Chapter 6, the following information is merely a summary of pain control using analgesics. Because pulpal and periapical pain involve inflammatory processes, the first choice of painkillers is nonsteroidal antiinflammatory drugs (NSAIDs).[69] However, no pain medication can replace the efficacy of thoroughly cleaning the root canal system to rid the tooth of the source of infection.[40]

Aspirin has been used as an analgesic for more than 100 years. In some cases, it may be more effective than 60 mg of codeine[21]; its analgesic and antipyretic effects are equal to those of acetaminophen, and its antiinflammatory effect is more potent.[24] However, aspirin's side effects include gastric distress, nausea, and gastrointestinal ulceration. In addition, its analgesic effect is inferior to that of ibuprofen, 400 mg. When NSAIDs and aspirin are contraindicated, such as in patients for whom gastrointestinal problems are a concern, acetaminophen is the preferred nonprescription analgesic. A recommended maximum daily dose of 4 g of acetaminophen is currently in force, and a further reduction of this dosage has been proposed to reduce the chance of acetaminophen-related liver toxicity.[68,101]

For moderate to severe pain relief, ibuprofen, an NSAID, is superior to aspirin (650 mg) and acetaminophen (600 mg) with or without codeine (60 mg). Also, ibuprofen has fewer side effects than the combinations with an opioid.[21,58] The maximal dose of 3.2 g in 24 hours should not be exceeded. Patients who take daily doses of aspirin for its cardioprotective benefit can take occasional doses of ibuprofen; however, it would be prudent to advise such patients to avoid regular doses of ibuprofen.[1] These patients would gain more relief by taking a selective cyclooxygenase (COX)-2 inhibitor, such as diclofenac or celecoxib.

Because of their antiinflammatory effect, NSAIDs can suppress swelling to a certain degree after surgical procedures. The good analgesic effect combined with the additional antiinflammatory benefit make NSAIDs, especially ibuprofen, the drug of choice for acute dental pain in the absence of any contraindication to their use. Ibuprofen has been used for more than 30 years and has been thoroughly evaluated.[24] If the NSAID alone does not have a satisfactory effect in controlling pain, then the addition of an opioid may provide additional analgesia. However, in addition to other possible side effects, opioids may cause nausea, constipation, lethargy, dizziness, and disorientation.

Laboratory Diagnostic Adjuncts

Chapter 15 discusses culturing techniques and indications. Because the results of culturing for anaerobic bacteria usually require at least 1 to 2 weeks, it is not considered routine in the management of an acute endodontic emergency. Thus, in an endodontic emergency, antibiotic treatment, *when indicated* (see Chapter 19), should begin immediately, because oral infections can progress rapidly.

Flare-Ups

An endodontic flare-up is defined as an acute exacerbation of a periradicular pathosis after the initiation or continuation of nonsurgical root canal treatment.[3] The incidence may be from 2% to 20% of cases.[56,78,84,117] A meta-analysis of the literature, using strict criteria, showed the flare-up frequency to be about 8.4%.[113] Endodontic flare-ups are more prevalent among females under the age of 20 years and may occur more in maxillary lateral incisors; in mandibular first molars, when there are large periapical lesions; and in the retreatment of previous root canals.[109] The presence of pretreatment pain may also be a predictor of potential posttreatment flare-ups.[56,110,117] Fortunately, there is no decrease in the endodontic success for cases that had a treatment flare-up.[60]

Endodontic flare-ups may occur for a variety of reasons, including preparation beyond the apical terminus, overinstrumentation, pushing dentinal and pulpal debris into the periapical area,[41] incomplete removal of pulp tissue, overextension of root canal filling material, chemical irritants (such as irrigants, intracanal medicaments, and sealers), hyper occlusion, root fractures, and microbiologic factors.[102] Although many of these cases can be pharmacologically managed (see Chapter 19), refractory cases may require periapical surgery, reentry into the tooth, the establishment of drainage either through the tooth or via trephination, or, at a minimum, adjustment of the occlusion.[21,97,102] The prophylactic use of antibiotics to decrease the incidence of flare-ups has been met with some controversy. Whereas earlier investigators[79] found that antibiotic administration before treatment of necrotic teeth reduced the incidence of flare-ups, more recent studies found antibiotic use either less effective than analgesics or not to affect in reducing interappointment emergencies or posttreatment symptoms.[94,110,117]

Cracked and Fractured Teeth

Described in detail in Chapters 1 and 22, cracks and incomplete fractures can be challenging to locate and diagnose, but their detection can be an essential component in the management of an acute dental emergency. In the early stages, cracks are small and difficult to discern. Removal of filling materials, applications of dye solutions, selective loading of cusps, transillumination, and magnification are helpful in their detection (Fig. 19.10). As the crack or fracture becomes more extensive, it can become easier to visualize. Because cracks are difficult to find and their symptoms can be so variable, the name *cracked tooth syndrome* has been suggested,[16] even though it is not indeed a syndrome. Cracks in vital teeth often exhibit a sudden and sharp pain, especially during mastication. Cracks in nonvital or obturated teeth tend to have more of a "dull ache" but can still be sensitive to mastication.

The determination of the presence of a crack or fracture is paramount because the prognosis for the tooth may be directly dependent on the extent of the crack or the longitudinal fracture. Management of cracks in vital teeth may be as simple as a bonded restoration or a full coverage crown. However, even the best efforts to manage a crack may be unsuccessful, often requiring endodontic treatment or extraction. Fractures in nonvital or obturated teeth may be more challenging. In addition, it must be determined whether the crack or fracture was the cause of pulpal necrosis and whether there has been an extensive periodontal breakdown. If so, the prognosis for the tooth is generally poor; thus, extraction is recommended (Fig. 19.11).

Summary

The management of endodontic emergencies is an essential part of dental practice. It can often be a disruptive part of the day for the clinician and staff, but it is an invaluable solution for the distressed patient. Methodical diagnosis and prognostic assessment are imperative, with the patient being informed of the various treatment alternatives.

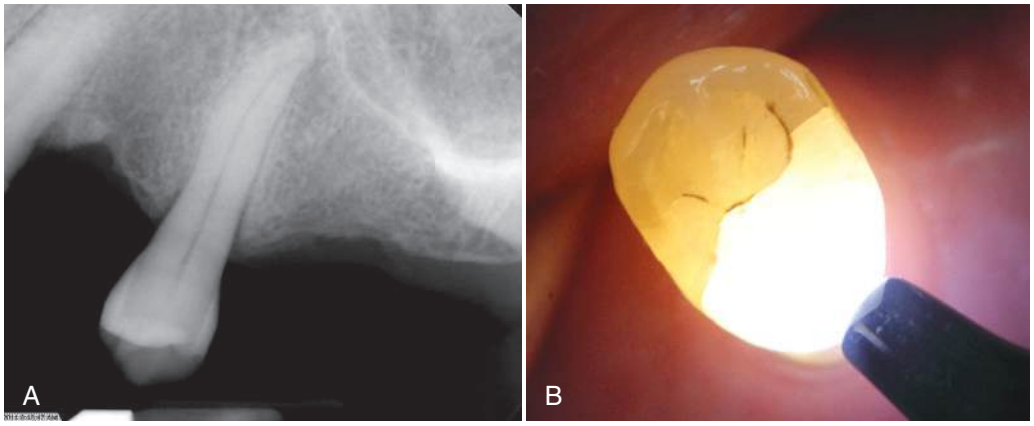


Fig. 19.10 **A,** Preoperative radiograph. **B,** Transillumination.

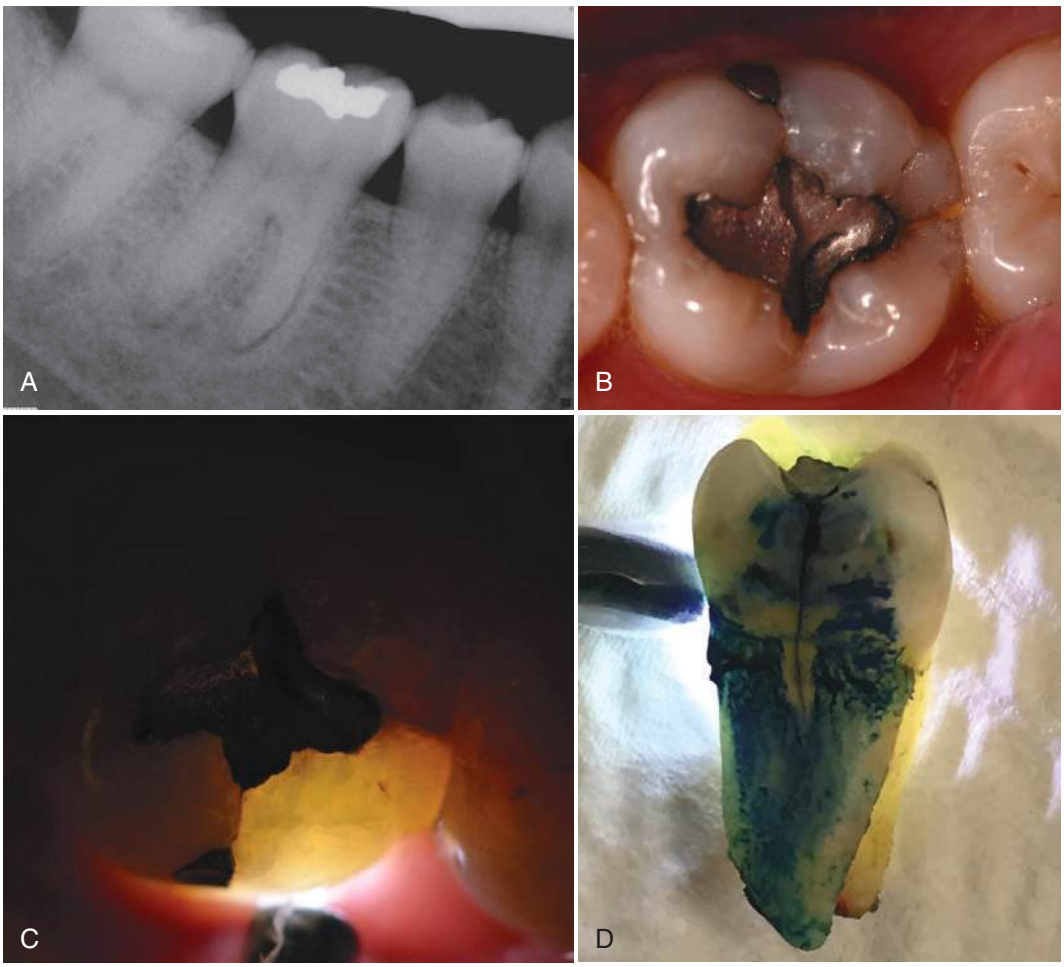


Fig. 19.11 **A,** Preoperative radiograph. **B,** Clinical view. **C,** Transillumination. **D,** Extracted tooth showing the extension of the stained longitudinal fracture.

References

1. Abramowicz M, editor: Do NSAIDs interfere with the cardioprotective effects of aspirin? *Med Lett Drugs Ther* 46:61, 2004.
2. Albahaireh ZS, Alnegrish AS: Postobturation pain after single and multiple-visit endodontic therapy: a prospective study, *J Dent* 26:227, 1998.
3. American Association of Endodontics: *Glossary of endodontic terms*, ed 7, Chicago, 2003, American Association of Endodontists.
4. Ashkenaz PJ: One-visit endodontics, *Dent Clin North Am* 28:853, 1984.
5. Asgary S, Hassanizadeh R, Torabzadeb H, et al: Treatment outcomes of 4 vital pulp therapies in mature molars, *J Endod* 44: 529–535, 2018.
6. Auslander WP: The acute apical abscess, *N Y State Dent J* 36:623, 1970.
7. Baumgartner JC, Xia T: Antibiotic susceptibility of bacteria associated with endodontic abscesses, *J Endod* 29:44, 2003.
8. Bence R, Meyers RD, Knoff RV: Evaluation of 5,000 endodontic treatment incidents of the open tooth, *Oral Surg Oral Med Oral Pathol* 49:82, 1980.
9. Bender IB: Pulpal pain diagnosis: a review, *J Endod* 26:175, 2000.
10. Bernal LA, Guillot E, Paquet C, et al: Beta-lactamase producing strains in the species *Prevotella intermedia* and *Prevotella nigrescens*, *Oral Microbiol Immunol* 13:36, 1998.
11. Beus H, Fowler S, Drum M, et al: What is the outcome of an incision and drainage procedure in endodontic patients? a prospective, randomized, single-blind study, *J Endod* 44:193–201, 2018.
12. Boutsioukis C, Psimma Z, Kastrinakis E: The effect of flow rate and agitation technique on irrigant extrusion *ex vivo*, *Int Endod J* 47:487, 2014.
13. Brook I: beta-Lactamase-producing bacteria in mixed infections, *Clin Microbiol Infect* 10:777, 2004.
14. Deleted in page review.
15. Bystrom A, Claesson R, Sundqvist G: The antibacterial effect of camphorated paramonochlorophenol, camphorated phenol and calcium hydroxide in the treatment of infected root canals, *Endod Dent Traumatol* 1:170, 1985.
16. Cameron CE: The cracked tooth syndrome, *J Am Dent Assoc* 93:971, 1976.
17. Carrotte P. Endodontics: part 3. Treatment of endodontic emergencies, *Br Dent J* 197:299, 2004.
18. Charara K, Friedman S, Sherman A, et al: Assessement of apical extrusion during root canal irrigation with the novel GentleWave system in a simulated apical environment, *J Endod* 42:135, 2016.
19. Chestner SB, Selman AJ, Friedman J, et al: Apical fenestration: solution to recalcitrant pain in root canal therapy, *J Am Dent Assoc* 77:846, 1986.
20. Chong BS, Pitt Ford TR: The role of intracanal medication in root canal treatment, *Int Endod J* 25:97, 1992.
21. Cooper SA, Beaver WT: A model to evaluate mild analgesics in oral surgery outpatients, *Clin Pharmacol Ther* 20:241, 1976.
22. Creech JH, Walton RE, Kaltenbach R: Effect of occlusal relief on endodontic pain, *J Am Dent Assoc* 109:64, 1984.
23. Desi P, Himel V: Comparative safety of various intracanal irrigation systems. *J Endod* 35:545, 2009.
24. Dionne RA, Phero JC, Becker DE: *Management of pain and anxiety in the dental office*, Philadelphia, 2002, Saunders.
25. Dorn SO, Moodnik RM, Feldman MJ, et al: Treatment of the endodontic emergency: a report based on a questionnaire—part I, *J Endod* 3:94, 1977.

26. Dorn SO, Moodnik RM, Feldman MJ, et al: Treatment of the endodontic emergency: a report based on a questionnaire—part II, *J Endod* 3:153, 1977.
27. Eleazer PD, Eleazer KR: Flare-up rate in pulpally necrotic molars in one-visit versus two-visit endodontic treatment, *J Endod* 24:614, 1998.
28. Field JW, Gutmann JL, Solomon ES, et al: A clinical radiographic retrospective assessment of the success rate of single-visit root canal treatment, *Int Endod J* 37:70, 2004.
29. Flynn TR: Anatomy of oral and maxillofacial infections. In Topazian RG, Goldberg MH, Hupp JR, editors: *Oral and maxillofacial infections*, ed 4, Philadelphia, 2002, WB Saunders, pp 188–213.
30. Fosse T, Madinier I, Hitzig C, et al: Prevalence of betalactamase-producing strains among 149 anaerobic gram negative rods isolated from periodontal pockets, *Oral Microbiol Immunol* 14:352, 1999.
31. Fouad AF, Rivera EM, Walton RE: Penicillin as a supplement in resolving the localized acute apical abscess, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 81:590, 1996.
32. Galani M, Tewari S, Sangwan P, et al: Comparative evaluation of postoperative pain and success rate after pulpotomy and root canal treatment in cariously exposed mature permanent molars - A randomized controlled trial, *J Endod* 43: 1953, 2017.
33. Gatchel RJ: Managing anxiety and pain during dental treatment, *J Am Dent Assoc* 123:37, 1992.
34. Gatewood RS, Himel VT, Dorn S: Treatment of the endodontic emergency: a decade later, *J Endod* 16:284, 1990.
35. Germack M, Sedgley CM, Sabbah W, et al: Antibiotic use in 2016 by members of the American Association of Endodontists: report of a national survey, *J Endod* 43:1615–1622, 2017.
36. Goldberg MH, Topazian RG: Odontogenic infections and deep fascial space infections of dental origin. In Topazian RG, Goldberg MH, Hupp JR, editors: *Oral and maxillofacial infections*, ed 4, Philadelphia, 2002, WB Saunders, pp 158–187.
37. Deleted in page review.
38. Grodinsky M, Holyoke EA: The fasciae and fascial spaces of the head, neck, and adjacent regions, *Am J Anat* 63:367, 1938.
39. Handal T, Olsen I, Walker CB, et al: Beta-lactamase production and antimicrobial susceptibility of subgingival bacteria from refractory periodontitis, *Oral Microbiol Immunol* 19:303, 2004.
40. Hargreaves KM, Keiser K: New advances in the management of endodontic pain emergencies, *J Calif Dent Assoc* 32:469, 2004.
41. Harrington GW, Natkin E: Midtreatment flare-ups, *Dent Clin North Am* 36:409, 1992.
42. Harrison JW: Irrigation of the root canal system, *Dent Clin North Am* 28:797, 1984.
43. Hasler JF, Mitchel DF: Analysis of 1628 cases of odontalgia: a corroborative study, *J Indianap Dist Dent Soc* 17:23, 1963.
44. Hasselgren G: Pains of dental origin, *Dent Clin North Am* 12:263, 2000.
45. Hasselgren G, Reit C: Emergency pulpotomy: pain relieving effect with and without the use of sedative dressings, *J Endod* 15:254, 1989.
46. Haug SR, Solfjeld AF, Ranheim LE, et al: Impact of case difficulty on endodontic mishaps in an undergraduate student clinic, *J Endod* 44:1088, 2018.
47. Hayward CMM, Griffin GE: Antibiotic resistance: the current position and the molecular mechanisms involved, *Br J Hosp Med* 52:473, 1994.
48. Hecht DW: Prevalence of antibiotic resistance in anaerobic bacteria: worrisome developments, *Clin Infect Dis* 39:92, 2004.
49. Hecht DW, Vedantam G, Osmolski JR: Antibiotic resistance among anaerobes: what does it mean? *Anaerobe* 5:421, 1999.
50. Heimdahl A, von Konow L, Satoh T, et al: Clinical appearance of orofacial infections of odontogenic origin in relation to microbiological findings, *J Clin Microbiol* 22:299, 1985.
51. Henry BM, Fraser JG: Trephination for acute pain management, *J Endod* 29:144, 2003.
52. Henry M, Reader A, Beck M: Effect of penicillin on postoperative endodontic pain and swelling in symptomatic necrotic teeth, *J Endod* 27:117, 2001.
53. Hohl TH, Whitacre RJ, Hooley JR, et al: *A self instructional guide: diagnosis and treatment of odontogenic infections*, Seattle, 1983, Stoma Press.
54. Holmes-Johnson E, Geboy M, Getka EJ: Behavior considerations, *Dent Clin North Am* 30:391, 1986.
55. Horrobin DF, Durnad LG, Manku MS: Prostaglandin E 1 modifies nerve conduction and interferes with local anesthetic action, *Prostaglandins* 14:103, 1997.
56. Imura N, Zuolo ML: Factors associated with endodontic flareups: a prospective study, *Int Endod J* 28:261, 1995.
57. Jacinto RC, Gomes BP, Ferraz CC, et al: Microbiological analysis of infected root canals from symptomatic and asymptomatic teeth with periapical periodontitis and the antimicrobial susceptibility of some isolated anaerobic bacteria, *Oral Microbiol Immunol* 18:285, 2003.
58. Jain AK, Ryan JR, McMahon G: Analgesic efficacy of low-dose ibuprofen in dental extraction, *Pharmacotherapy* 6:318, 1986.
59. Keenan JV, Farman AG, Fedorowica Z, et al: A cochrane systematic review finds no evidence to support the use of antibiotics for pain relief in irreversible pulpitis, *J Endod* 32:87, 2006.
60. Kerekes K, Tronstad L: Long-term results of endodontic treatment performed with a standardized technique, *J Endod* 5:83, 1979.
61. Khemaleelakul S, Baumgartner JC, Pruksakorn S: Identification of bacteria in acute endodontic infections and their antimicrobial susceptibility, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 94:746, 2002.
62. Kim E, Lee SJ: Electronic apex locator [Review], *Dent Clin North Am* 48:35, 2004.
63. Kuriyama T, Karasawa T, Nakagawa K, et al: Bacteriologic features and antimicrobial susceptibility in isolates from orofacial odontogenic infections, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 90:600, 2000.
64. Kuriyama T, Karasawa T, Nakagawa K, et al: Incidence of beta-lactamase production and antimicrobial susceptibility of anaerobic gram-negative rods isolated from pus specimens of orofacial odontogenic infections, *Oral Microbiol Immunol* 16:10, 2001.
65. Kuriyama T, Williams DW, Yanagisawa M, et al: Antimicrobial susceptibility of 800 anaerobic isolates from patients with dentoalveolar infection to 13 oral antibiotics, *Oral Microbiol Immunol* 22:285, 2007.
66. Kvist T, Molander A, Dahlen G, et al: Microbiological evaluation of one- and two-visit endodontic treatment of teeth with apical periodontitis: a randomized, clinical trial, *J Endod* 30:572, 2004.
67. Lakhassassi N, Elhajoui N, Lodter JP, et al: Antimicrobial susceptibility variation of 50 anaerobic periopathogens in aggressive periodontitis: an interindividual variability study, *Oral Microbiol Immunol* 20:244, 2005.
68. Larson AM, Polson J, Fontana RJ, et al: Acetaminophen-induced acute liver failure: results of a United States multicenter, prospective study, *Hepatology* 42:1364, 2005.
69. Lee M, Winkler J, Hartwell G, et al: Current trends in endodontic practice: emergency treatments and technological armamentarium, *J Endod* 35:35, 2009.
70. Levitt GW: The surgical treatment of deep neck infections, *Laryngoscope* 81:403, 1970.
71. Lipton JA, Ship JA, Larach-Robinson D: Estimated prevalence and distribution of reported orofacial pain in the United States, *J Am Dent Assoc* 124:115, 1993.
72. Madigan MT, Martinko JM, Parker J: *Brock biology of microorganisms*, ed 9, Upper Saddle River, NJ, 2000, Prentice-Hall.
73. Matusow RJ, Goodall LB: Anaerobic isolates in primary pulpal–alveolar cellulitis cases: endodontic resolutions and drug therapy considerations, *J Endod* 9:535, 1983.
74. Metka ME, Liem VML, Parsa A, et al: Cone-beam computed tomographic scans in comparison with periapical radiographs for root canal length measurement: an in situ study, *J Endod* 40:1206, 2014.
75. Mitchell RP, Yang S, Baumgartner JC: Comparison of apical extrusion of NaOCl using the EndoVac or needle irrigation of root canals, *J Endod* 36:338, 2010.
76. Deleted in page review.
77. Moos HL, Bramwell JD, Roahen JO: A comparison of pulpectomy alone versus pulpectomy with trephination for the relief of pain, *J Endod* 22:422, 1996.
78. Morse DR, Koren LZ, Esposito JV, et al: Asymptomatic teeth with necrotic pulps and associated periapical radiolucencies: relationship of flare-ups to endodontic instrumentation, antibiotic usage and stress in three separate practices at three different time periods, *Int J Psychosom* 33:5, 1986.
79. Morse DR, Furst ML, Belott RM, et al: Infectious flare-ups and serious sequelae following endodontic treatment: a prospective randomized trial on efficacy of antibiotic prophylaxis in cases of asymptomatic pulpal–periapical lesion, *Oral Surg* 64:96, 1987.

80. Nagle D, Reader A, Beck M, et al: Effect of systemic penicillin on pain in untreated irreversible pulpitis, *Oral Surg Oral Med Oral Pathol* 90:636, 2000.
81. Natkin E: Treatment of endodontic emergencies, *Dent Clin North Am* 18:243, 1974.
82. Nusstein J, Reader A, Nist R, et al: Anesthetic efficacy of the supplemental intraosseous injection, *J Endod* 24:487, 1998.
83. Nyerere JW, Matee MI, Simon EN: Emergency pulpotomy in relieving acute dental pain among Tanzanian patients, *BMC Oral Health* 6:1, 2006.
84. Oginni AO, Udoye CI: Endodontic flare-ups: comparison of incidence between single and multiple visit procedures in patients attending a Nigerian teaching hospital, *BMC Oral Health* 4:4, 2004.
85. Ogundiya DA, Keith DA, Mirowski J: Cavernous sinus thrombosis and blindness as complications of an odontogenic infection, *Oral Maxillofac Surg* 47:1317, 1989.
86. Oliet S: Single-visit endodontics: a clinical study, *J Endod* 24:614, 1998.
87. Pallasch TJ: Antibiotics in endodontics, *Dent Clin North Am* 23:737, 1979.
88. Pallasch TJ: Pharmacokinetic principles of antimicrobial therapy, *Periodontol 2000* 10:5, 1996.
89. Pallasch TJ, Slots J: Antibiotic prophylaxis and the medically compromised patient, *Periodontol 2000* 10:107, 1996.
90. Patel R: Clinical impact of vancomycin-resistant enterococci, *J Antimicrob Chemother* 51(Suppl 3):iii13, 2003.
91. Pekruhn RB: The incidence of failure following single-visit endodontic therapy, *J Endod* 12:68, 1986.
92. Penesis VA, Fitzgerald PI, Fayad MI, et al: Outcome of one-visit and two-visit endodontic treatment of necrotic teeth with apical periodontitis: a randomized controlled trial with one-year evaluation, *J Endod* 34:251, 2008.
93. Peters LB, Wesselink PR: Periapical healing of endodontically treated teeth in one and two visits obturated in the presence or absence of detectable microorganisms, *Int Endod J* 35:660, 2002.
94. Pickenpaugh L, Reader A, Beck M, et al: Effect of prophylactic amoxicillin on endodontic flare-up in asymptomatic, necrotic teeth, *J Endod* 27:53, 2001.
95. Reddy SA, Hicks ML: Apical extrusion of debris using two hand and two rotary instrumentation techniques, *J Endod* 24:180, 1998.
96. Roane JB, Dryden JA, Grimes EW: Incidence of postoperative pain after single- and multiple-visit endodontic procedures, *Oral Surg Oral Med Oral Pathol* 55:68, 1983.
97. Rosenberg PA, Babick PJ, Schertzer L, et al: The effect of occlusal reduction on pain after endodontic instrumentation, *J Endod* 24:492, 1998.
98. Rudner WL, Oliet S: Single-visit endodontics: a concept and a clinical study, *Compend Contin Educ Dent* 2:63, 1981.
99. Rugh JD: Psychological components of pain, *Dent Clin North Am* 31:579, 1987.
100. Sandor GK, Low DE, Judd PL, et al: Antimicrobial treatment options in the management of odontogenic infections, *J Can Dent Assoc* 64:508, 1998. Comment in *J Can Dent Assoc* 65:602, 1999.
101. Schilling A, Corey R, Leonard M, et al: Acetaminophen: old drug, new warnings, *Cleve Clin J Med* 77:19, 2010, doi: 10.3949/ccjm.77a.09084.
102. Seltzer S, Naidorf IJ: Flare-ups in endodontics. 1. Etiological factors, *J Endod* 11:472, 1985.
103. Simon JH, Chimenti RA, Mintz GA: Clinical significance of the pulse granuloma, *J Endod* 8:116, 1982.
104. Siqueira JF, Rocas IN: Microbial causes of endodontic flareups, *Int Endod J* 36:433, 2003.
105. Sjogren U, Figdor D, Persson S, et al: Influence of infection at the time of root filling on the outcome of endodontic treatment of teeth with apical periodontitis, *Int Endod J* 30:297, 1997.
106. Southard DW, Rooney TP: Effective one-visit therapy for the acute periapical abscess, *J Endod* 10:580, 1984.
107. Sutherland S, Matthews DC. Emergency management of acute apical periodontitis in the permanent dentition: a systematic review of the literature, *J Can Dent Assoc* 69:160, 2003.
108. Tendolkar PM, Baghdayan AS, Shankar N: Pathogenic enterococci: new developments in the 21st century, *Cell Mol Life Sci* 60:2622, 2003.
109. Torabinejad M, Dorn SO, Eleazer PD, et al: The effectiveness of various medications on postoperative pain following root canal obturation, *J Endod* 20:427, 1994.
110. Torabinejad M, Kettering JD, McGraw JC, et al: Factors associated with endodontic interappointment emergencies of teeth with necrotic pulps, *J Endod* 14:261, 1988.
111. Torabinejad M, Walton R: *Endodontics: principles and practice*, ed 4, St. Louis, 2009, Saunders.
112. Trope ME, Delano EO, Orstavik D: Endodontic treatment of teeth with apical periodontitis: single vs. multivisit treatment, *J Endod* 25:345, 1999.
113. Tsesis I, Faivishevsky V, Fuss Z, et al: Flare-ups after endodontic treatment: a meta-analysis of literature, *J Endod* 34:1177, 2008.
114. Turkun M, Cengiz T: The effects of sodium hypochlorite and calcium hydroxide in tissue dissolution and root canal cleanliness, *Int Endod J* 30:335, 1997.
115. van Winkelhoff AJ, Winkel EG, Barendregt D, et al: beta-Lactamase producing bacteria in adult periodontitis, *J Clin Periodontol* 24:538, 1997.
116. Vera J, Ochoa J, Romero M, et al, Intracanal cryotherapy reduces postoperative pain in teeth with symptomatic apical periodontitis: A randomized multicenter clinical trial, *J Endod* 44:4, 2018.
117. Walton R, Fouad A: Endodontic interappointment flare-ups: a prospective study of incidence and related factors, *J Endod* 18:172, 1992.
118. Weiger R, Axmann-Krcmar D, Löst C: Prognosis of conventional root canal treatment reconsidered, *Endod Dent Traumatol* 14:1, 1998.
119. Weiger R, Rosendahl R, Lost C: Influence of calcium hydroxide intracanal dressings on the prognosis of teeth with endodontically induced periapical lesions, *Int Endod J* 33:219, 2000.
120. Weine FS, Healey HJ, Theiss EP: Endodontic emergency dilemma: leave tooth open or keep it closed? *Oral Surg Oral Med Oral Pathol* 40:531, 1975.
121. Whitney CG, Farley MM, Hadler J, et al: Increasing prevalence of multidrug-resistant Streptococcus pneumoniae in the United States, *N Engl J Med* 343:1917, 2000.
122. Yun MW, Hwang CF, Lui CC: Cavernous sinus thrombus following odontogenic and cervicofacial infection, *Eur Arch Otorhinolaryngol* 248:422, 1991.